

Measured CPU / CUDA correctness and acceleration

Case	CPU s	CUDA s	CPU/CUDA	Equivalent	Physical pass
annular-256x64	0.24450	0.08799	2.779	True	True
annular-512x128	1.04508	0.22076	4.734	True	True
annular-1024x256	5.42082	0.74338	7.292	True	True
poiseuille-72x24	5.18851	12.87225	0.403	True	True
taylor-green-64	1.73491	1.17926	1.471	True	True
abc-24-cubed	0.63635	0.50624	1.257	True	True
advected-shear-32	1.25692	1.66117	0.757	True	True
wall-law-batch	0.72737	0.38401	1.894	True	None
SA-128x112-five-iterations	118.80939	17.17528	6.917	False	None
TensorLBM-shear-64	0.23077	0.15127	1.525	True	True

Float64; one CPU thread; actual algorithm warm-up; three synchronized repeated executions.

Runtime includes setup/diagnostics/raw host export, excludes plots.

Annular CPU and CUDA use identical torch CSR/Jacobi algorithms.

SA five-iteration comparison is not a converged physical benchmark.

Wall-law throughput is constitutive inversion, not a complete CFD solve.

Allocated torch GPU peak is not whole-device memory. All slower CUDA results retained.

